

Aqua Solutions Final Report

Open-source data collection for water
quality monitoring in Harare, Zimbabwe

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Initial Problem, Design, and Difficulties

Problem Specification:

- 80% of water consumption in Harare, Zimbabwe is from unregulated boreholes.
- Faecal contamination leads to waterborne disease outbreaks.

Allen Chafa's Original Design:

- Fuzzy logic control system:
 - Sensor input → Chlorine injection output

Difficulties:

- Chlorine dosing is a high stakes output:
 - Too much is toxic.
 - Too little is ineffective.
- Proxies were insufficient to predict contamination.
- Algorithm collapsed on single sensor failure.



Our Solution: A report-labelled water safety pipeline



COMPONENT 1

Sensor Node

Low-cost field nodes measure pH, turbidity, ORP, UV, and Chlorine Residual.



COMPONENT 2

Reporting Unit

Website: Village Health Workers (VHWs) log illness reports from a dashboard at a clinic/office.

SMS: VHWs can report directly, or community members can text the station.



COMPONENT 3

ML Pipeline

XGBoost model is pre-trained on synthetic data and ready to retrain on real readings.

C code export enables direct deployment to the sensor node.



Benefits



Report from anywhere

Dashboard or SMS – anyone can report without visiting a clinic or using a smartphone



No Lab Expenses

Illness reports generate immediate labels, replacing expensive and slow lab water testing



Immediate offline inference

Once trained, inference runs directly on the sensor node offline and immediately flags risk.



Open source, ready to hand over

Open-source and designed for handover to Allen Chafa and his team





Step 1: Testing the Predictive Ceiling

full_dataset.csv

- 25,000 readings from South African boreholes
 - pH, turbidity, temperature, E. coli
- Macro-F1 readings
 - Always predict high = 0.317
 - XGBoost = 0.324
- Severe class imbalance (95% high-risk labels) limits model discriminability.

Water_potability.csv

- 9 chemistry features with a binary potability label
- Macro-F1 readings
 - Always predict high = 0.379
 - Logistic regression = 0.495
 - XGBoost = 0.598
- Dataset of uncertain origin (kaggle) with no explanation for the potability label

Conclusion

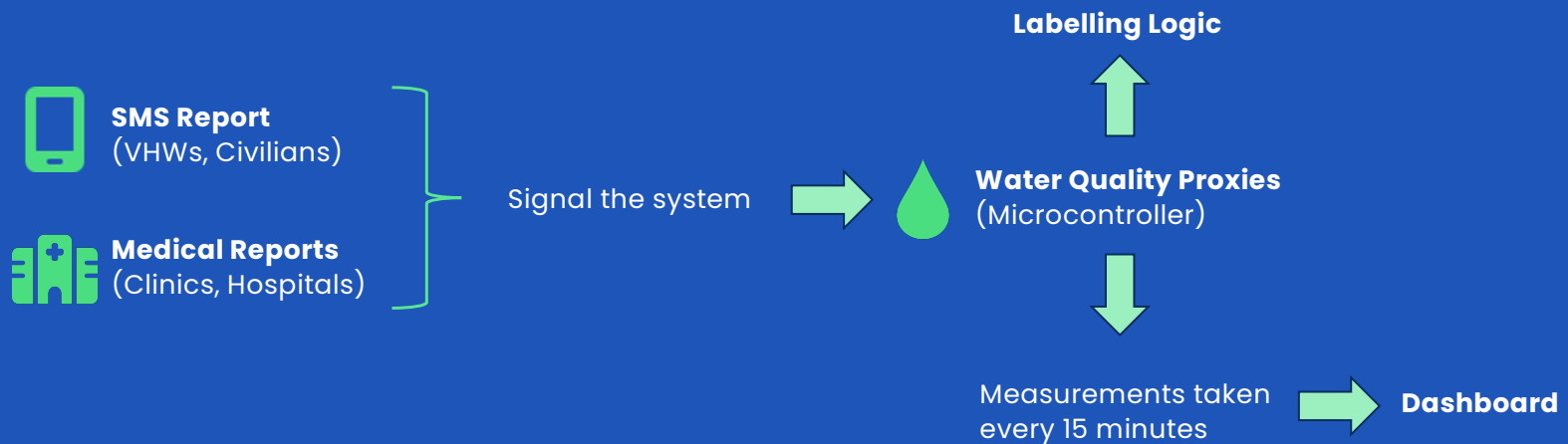
The method works, the datasets don't!



The Dataset Problem

- **Missing Values**
- **Bacterial Correlation vs. Contamination → Labelling:** How do we know when water is actually contaminated?
- **Inappropriate Data:** Not all water sources have the same contamination mechanisms.
- **Skewed Results:** High density of HIGH risk water datapoints, reducing the model's trained inference capability.

New Data Pipeline



Labelling Logic



Confidence tiers

Tiers	Label	Weights
<0.5	Safe	0.1
0.5 - 1.5	Risky	0.3
1.5 - 2.5	Risky	0.6
2.5 <	Risky	0.9



Labelling design choice justification



7 Day Window

WHO Cholera incubation up to 5 days. Adding a 2 day reporting window, gives a 7 day rolling window.



Logarithmic decay

Most cases present within 48 hours, so recent reports should carry more weight than older ones.



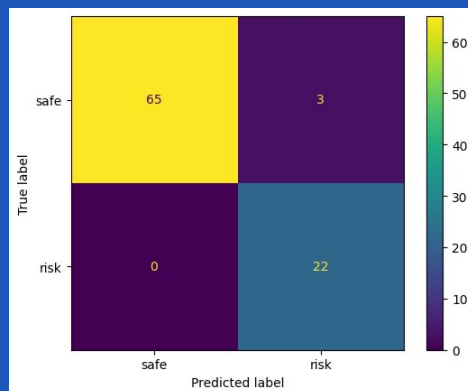
Confidence tiers

- Sustained reporting, increases contamination likelihood.
- No reports $\neq$ safe water. Low weighting used.
- Discrete tiers for transparency and easy to update as understanding improves.
- Note – thresholds are arbitrary and need calibration as data comes in.
- Specifically, the 0.5 safe threshold is designed to filter out background illness reports.



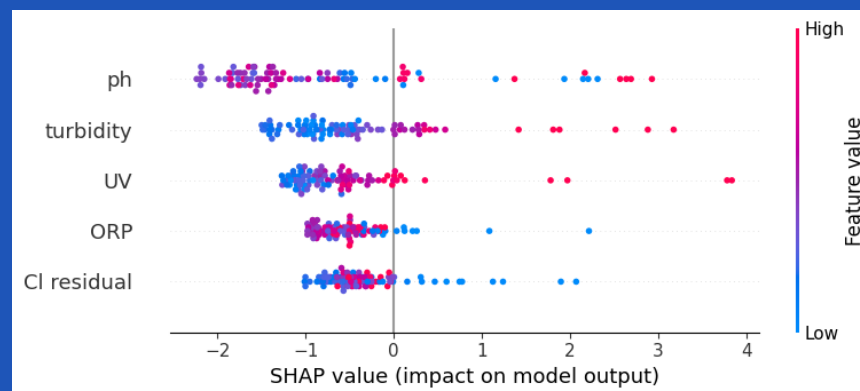
ML Pipeline

- XGBoost pipeline trained on synthetic data, ready to retrain when field data accumulates
- Model exported to C via m2cgen for offline inference on the sensor node



Confusion Matrix

Near-perfect classification on synthetic test data



SHAP Feature Importance

Shows how each feature pushes the model towards a decision (positive SHAP indicates a 'risky' label)

User Interface

DHIS2: District Health Information Software

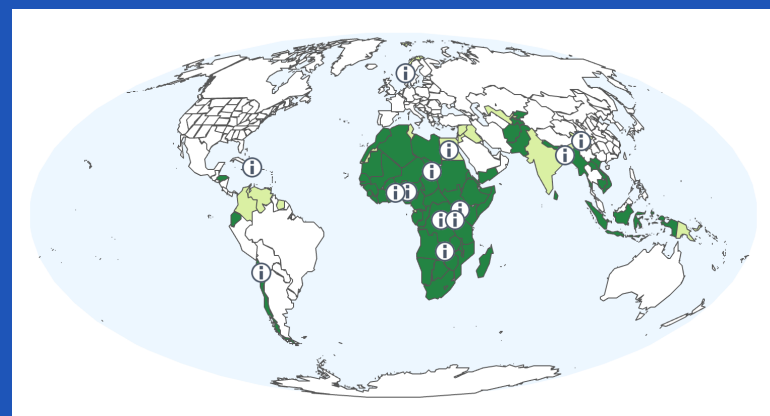
- Open-source software developed by Microsoft for the collection, analysis, presentation, and sharing of aggregate and individual-level data.
- Sub-apps (e.g. Data Visualizer, Maintenance, Users) are used to configure dashboards and a reporting system.

Objectives:

- Direct to the point (e.g. VHWs are routed directly to medical form upon sign-in)
- Easy to navigate
- High risk events “jump out” at the user

Why DHIS2?

- Widely used, tried and tested
- Many VHWs already trained to use it
- Incorporates SMS capabilities and offline use



Previous Government Dashboard

CHITESHI CHIMIRO · 32 ZVITESHI · 7-ZUVA HWINDO				Nharaunda dzese		
STN-1 · Avenues — central clinic	2026-06-01T17:55 · pH 6.9 · 12 NTU · 23.0°C · 0.0 mm · Cl 0.94 mg/L · ORP 345 mV · UV ₂₅₄ 0.153	ZVAKANAKA	Close	Muenzaniso	Medical	
STN-2 · Belvedere — community hall	2026-06-01T17:55 · pH 6.7 · 14 NTU · 22.0°C · 0.0 mm · Cl 0.31 mg/L · ORP 607 mV · UV ₂₅₄ 0.085	KUSINA KUCHENGETEKA	Close	Muenzaniso	Medical	
STN-3 · Eastlea — primary school	2026-06-01T17:55 · pH 7.6 · 9 NTU · 23.9°C · 0.0 mm · Cl 0.77 mg/L · ORP 524 mV · UV ₂₅₄ 0.126	KUSINA KUCHENGETEKA	Close	Muenzaniso	Medical	
STN-4 · Mbare — Musika market	2026-06-01T17:55 · pH 7.3 · 6 NTU · 23.8°C · 0.0 mm · Cl 0.46 mg/L · ORP 728 mV · UV ₂₅₄ 0.107	ZVAKANAKA	Close	Muenzaniso	Medical	
STN-5 · Hatfield — community borehole	2026-06-01T17:55 · pH 7.2 · 7 NTU · 24.4°C · 0.0 mm · Cl 0.74 mg/L · ORP 561 mV · UV ₂₅₄ 0.118	ZVAKANAKA	Close	Muenzaniso	Medical	
STN-6 · Newlands — shopping centre	2026-06-01T17:55 · pH 6.8 · 10 NTU · 22.9°C · 0.0 mm · Cl 0.75 mg/L · ORP 654 mV · UV ₂₅₄ 0.044	KUSINA KUCHENGETEKA	Close	Muenzaniso	Medical	
STN-7 · Milton Park — health post	2026-06-01T17:55 · pH 7.1 · 2 NTU · 21.1°C · 0.0 mm · Cl 0.63 mg/L · ORP 793 mV · UV ₂₅₄ 0.105	KUSINA KUCHENGETEKA	Close	Muenzaniso	Medical	
STN-8 · Hillside — water point	2026-06-01T17:55 · pH 7.6 · 0 NTU · 22.9°C · 0.0 mm · Cl 0.30 mg/L · ORP 615 mV · UV ₂₅₄ 0.000	ZVAKANAKA	Close	Muenzaniso	Medical	
STN-9 · Mt Pleasant — north well	2026-06-01T17:55 · pH 7.2 · 6 NTU · 23.0°C · 0.0 mm · Cl 0.17 mg/L · ORP 480 mV · UV ₂₅₄ 0.102	KUSINA KUCHENGETEKA	Close	Muenzaniso	Medical	
STN-10 · Greendale — east settlement	2026-06-01T17:55 · pH 7.1 · 4 NTU · 21.1°C · 0.0 mm · Cl 0.33 mg/L · ORP 659 mV · UV ₂₅₄ 0.109	ZVAKANAKA	Close	Muenzaniso	Medical	
STN-11 · Causeway — government complex	2026-06-01T17:55 · pH 7.3 · 3 NTU · 27.0°C · 0.0 mm · Cl 0.33 mg/L · ORP 627 mV · UV ₂₅₄ 0.159	ZVAKANAKA	Close	Muenzaniso	Medical	
STN-12 · Kopje — civic hall	2026-06-01T17:55 · pH 6.8 · 15 NTU · 21.6°C · 0.0 mm · Cl 0.48 mg/L · ORP 510 mV · UV ₂₅₄ 0.130	ZVAKANAKA	Close	Muenzaniso	Medical	
STN-13 · CBD — central market	2026-06-01T17:55 · pH 7.8 · 14 NTU · 19.9°C · 8.1 mm · Cl 0.93 mg/L · ORP 623 mV · UV ₂₅₄ 0.059	ZVAKANAKA	Close	Muenzaniso	Medical	
STN-14 · Africa Unity Square — fountain	2026-06-01T17:55 · pH 7.4 · 5 NTU · 23.3°C · 0.0 mm · Cl 0.17 mg/L · ORP 594 mV · UV ₂₅₄ 0.048	ZVAKANAKA	Close	Muenzaniso	Medical	
STN-15 · Workington — industrial water	2026-06-01T17:55 · pH 7.1 · 3 NTU · 23.4°C · 0.0 mm · Cl 0.39 mg/L · ORP 668 mV · UV ₂₅₄ 0.090	KUSINA KUCHENGETEKA	Close	Muenzaniso	Medical	
STN-16 · Avondale — north clinic	2026-06-01T17:55 · pH 7.2 · 8 NTU · 22.4°C · 0.0 mm · Cl 0.56 mg/L · ORP 819 mV · UV ₂₅₄ 0.054	ZVAKANAKA	Close	Muenzaniso	Medical	
STN-17 · Belgravia — community well	2026-06-01T17:55 · pH 7.4 · 4 NTU · 20.7°C · 0.0 mm · Cl 0.32 mg/L · ORP 761 mV · UV ₂₅₄ 0.156	ZVAKANAKA	Close	Muenzaniso	Medical	

MISHUMO YECHIRWERE CHAZVINO UNO (6) · RISK TIER			
2026-06-01 13:02 · STN-15 · Workington — industrial water	Isina kukwana · hapana data yakarongeka iripo 83 yakanyorwa		
2026-06-01 08:56 · STN-9 · Mt Pleasant — north well	ZVAKANYANYA 1354 yakanyorwa		
2026-06-01 08:40 · STN-3 · Eastlea — primary school		HIGH	
2026-06-01 08:39 · STN-7 · Milton Park — health post		MEDIUM	
2026-06-01 08:38 · STN-6 · Newlands — shopping centre	Isina kukwana · hapana data yakarongeka iripo 1230 yakanyorwa		
2026-05-31 12:07 · STN-2 · Belvedere — community hall	Isina kukwana · hapana data yakarongeka iripo 16 yakanyorwa		

Medical Reports

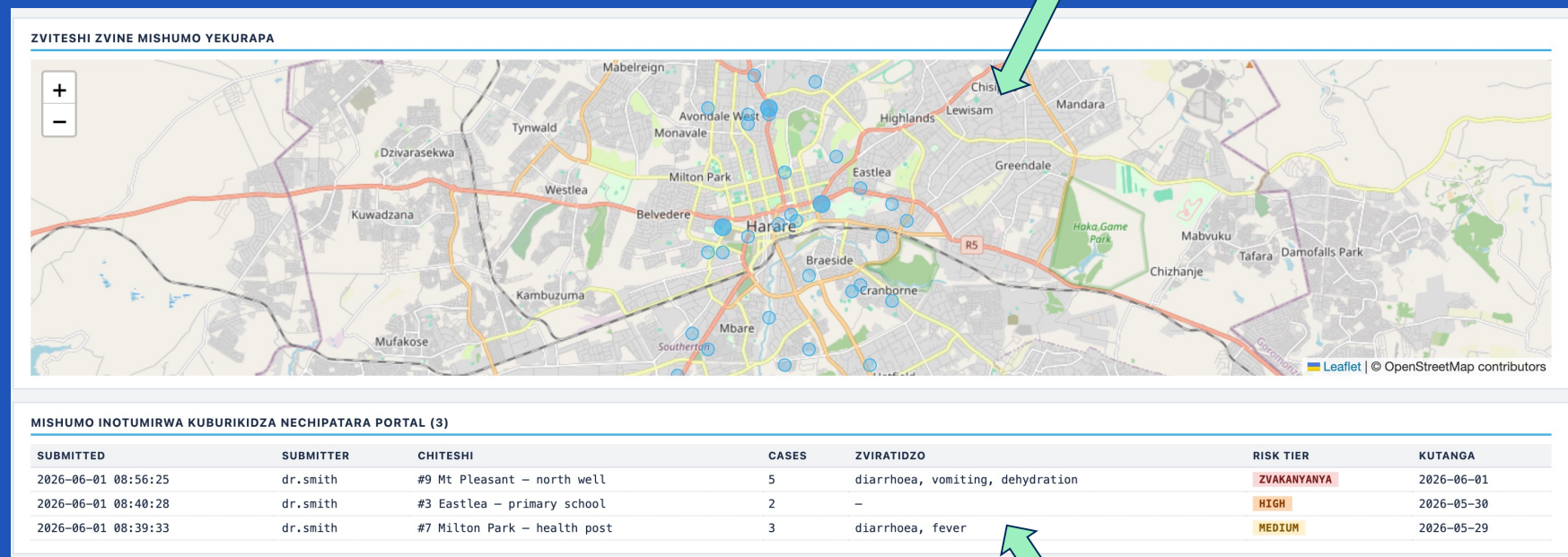
Boreholes

Water Quality Status

Governance Delegation Tasks

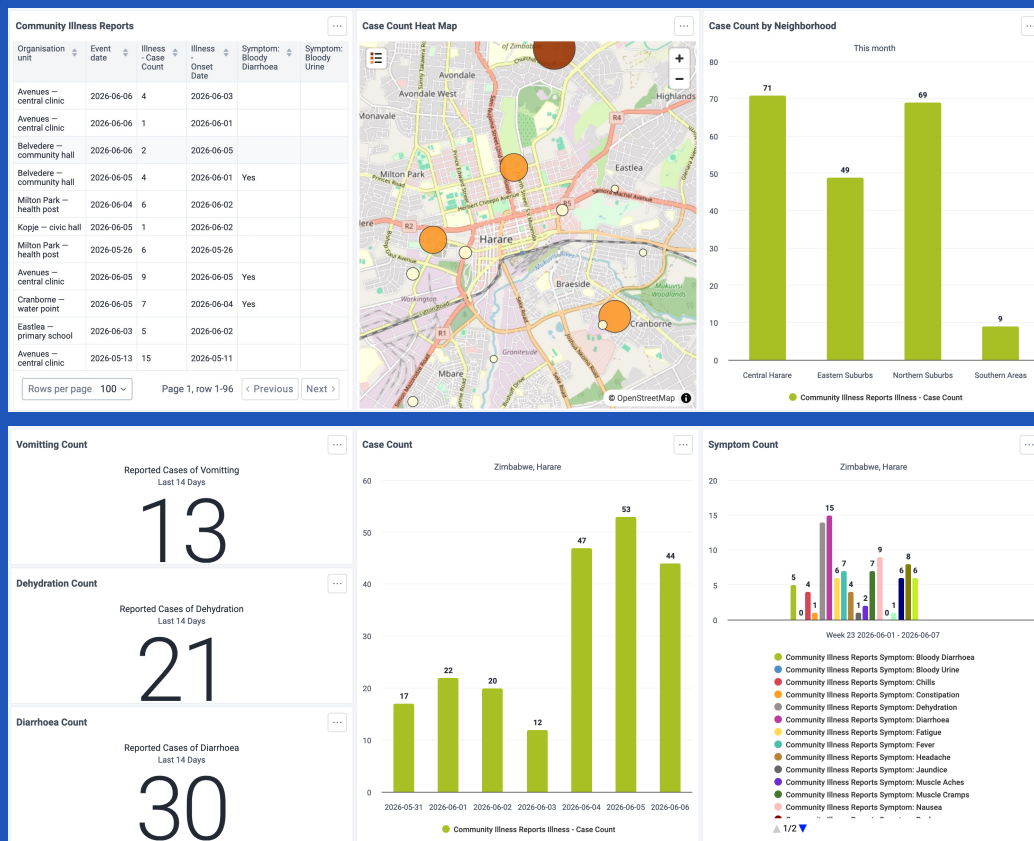
Previous Medical Dashboard (for VHWs)

Map of Harare's Boreholes



Previous Medical Reports

DHIS2 Government Dashboard



Key Elements:

- Dataset
- Heatmap of borehole case reports
- Case count by region of Harare
- Case count by cholera-indicative symptoms (vomiting, fever, diarrhoea)

Changes:

- Data visualization
- Key metrics draw attention (heatmap, case count values)
- Heat map presented to government officials. VHVs have knowledge of their neighborhood's well-being.

DHIS2 Medical Dashboard

Report date *	yyyy-mm-dd	×
Organisation unit *	Avenues — central clinic ×	
Illness - Onset Date * ⓘ	yyyy-mm-dd	×
Illness - Case Count * ⓘ		
Symptom: Diarrhoea	☐	Yes
Symptom: Vomiting	☐	Yes
Symptom: Fever	☐	Yes
Symptom: Dehydration	☐	Yes
Symptom: Weight Loss	☐	Yes
Symptom: Muscle Cramps	☐	Yes
Symptom: Shock	☐	Yes

Key Elements:

- Medical Form
- Records:
 - 18 symptoms across most common waterborne diseases in Zimbabwe
 - Case count
 - Symptom onset date (used for labeling)
- Each medical professional is linked to a clinic/region when setting up their profile



Next Steps

- **Field Deployment:**
 - Build sensor node with proposed sensor changes
 - Collect data and train an ML algorithm (XGBoost framework in place)
 - Validate the core assumption → Are illness-reports are a strong enough contamination proxy? Despite noise from recontamination and non-water-borne illnesses.
 - **Full ML Consolidation:**
 - Integrate labelling logic and data integration into the ML Model
 - Recalibrate thresholds on real data
 - **DHIS2 Consolidation:** Resolve discrepancies between Railway and DHIS2 versions
 - **Additional Analysis:**
 - Addition of rainfall/temporal features
 - Data cleaning mechanisms - pre-labelling some drift examples
 - Disease-symptom relationships
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Thank you!

